

PHYS1408-002 Exam I September 25, 2015

$$\mathbf{x}_f = \mathbf{x}_i + \mathbf{v}_{xi}t + \frac{1}{2}\mathbf{a}_x t^2$$
$$\mathbf{x}_f = \mathbf{x}_i + \frac{\mathbf{v}_{xf} + \mathbf{v}_{xi}}{2}t$$

$$\mathbf{v}_{xf} = \mathbf{v}_{xi} + \mathbf{a}_x t$$
$$\mathbf{v}_{xf}^2 = \mathbf{v}_{xi}^2 + 2\mathbf{a}_x(\mathbf{x}_f - \mathbf{x}_i)$$

1. Is the following a scientific hypothesis? **“Space is filled with an undetectable substance.”**

- a) yes, it is a valid scientific hypothesis
- b) no it is not, because it is not true
- c) no it is not, because it is not testable
- d) no it is not, because it is not convincing
- e) no it is not, because it cannot be proven to be true

2. Is the following a scientific hypothesis? **“Atoms are the smallest units of matter.”**

- a) yes, it is a valid scientific hypothesis
- b) no it is not, because it is not true
- c) no it is not, because it is not testable
- d) no it is not, because it is not convincing
- e) no it is not, because it cannot be proven to be true

3. The scientific method requires that you construct something based on your observations and test it. What must be constructed and tested?

- a) table top model b) alibi c) explanation d) mental model e) procedure

4. Aunt Matilda goes to a well and throws a penny straight down the well at 3 m/s. She hears a splash after 1.2 seconds. How deep is the well?

- a) 3.5 m b) 7.1 m c) 9.5 m d) 10.6 m e) 11.8 m

5. A ball is thrown upward. As it passes 7 m height it is traveling at 2 m/s up. What was its initial upward velocity?

- a) 11.1 m/s b) 11.3 m/s c) 11.5 m/s d) 11.7 m/s e) 11.9 m/s

6. A plane is coming in to land on a carrier deck 150 m long. If it lands at 50 m/s and decelerates smoothly (it doesn't really, but for this problem assume), what must its constant acceleration be?

- a) -8.0 m/s² b) -8.1 m/s² c) -8.3 m/s² d) -8.5 m/s² e) -8.7 m/s²

7. Given the vectors $\vec{A} = 3\hat{i} + 2\hat{j}$ and $\vec{B} = 1\hat{i} + 2\hat{j}$. What is the angle between $\vec{A} + 2\vec{B}$ and the positive x-axis?

- a) 35° b) 40° c) 45° d) 50° e) 55°

8. A block with mass 20 kg is sliding up a plane (coefficient of friction 0.25, inclined at 15°) at a speed of 2 m/s to the right (positive x-direction). How far does it go up along the plane before it comes to rest momentarily?

- a) 0.21 m b) 0.41 m c) 0.57 m d) 0.78 m e) 0.95 m

9. A mass (10 kg) rests on a surface ($\mu_{\text{static}}=0.5$) connected by a cord to a hanging mass of 4 kg. The connecting cord is mass-less and stretch-free. The pulley is mass-less and frictionless. What is the **force of friction** on the heavier mass?

- a) 19.6 kg m/s² b) 29.4 kg m/s² c) 39.2 kg m/s² d) 49.0 kg m/s² e) 58.8 kg m/s²

10. A ball rolls off a table of height 0.85 m at a speed of 3 m/s horizontal. When it hits the ground, at what angle (below horizontal) does it hit?

- a) 54° b) 57° c) 60° d) 63° e) 65°

11. A cannon is fired at 30° above the horizontal with a muzzle velocity of 25 m/s. The cannon is firing from ground level and the field is level. How far down field does the cannon ball strike the ground?

- a) 55 m b) 63 m c) 79 m d) 86 m e) 99 m

12. A car of mass 1000 kg turns a 10° inclined corner of radius 100 m that has a coefficient of friction of 0.4. What is the maximum speed the car can have around this corner without sliding up the incline?

- a) 17 m/s b) 21 m/s c) 26 m/s d) 31 m/s e) 35 m/s

13. A block of mass 2 kg is sliding down an incline that has a coefficient of kinetic friction of 0.4. The block slows from 2 m/s initially to rest. The incline is at 30° . What was the magnitude of the force of friction during the time the block slowed?

- a) 3.4 kg m/s^2 b) 4.0 kg m/s^2 c) 5.1 kg m/s^2 d) 6.8 kg m/s^2 e) 8.3 kg m/s^2

14. Mass 5 kg is resting on the table. Mass 2 kg is hanging off the side of the table as shown. They are connected by a cord over a pulley. The surface is frictionless. What is the tension in the string when the masses are released?

- a) 14 N
b) 16 N
c) 20 N
d) 26 N
e) 34 N

15. A wooden block (mass 0.05 kg) is set on a rotating record. The mass is 0.1 m from the axis of rotation and is moving with the disk at 0.4 m/s. What minimum coefficient of friction is required to ensure the mass does not slide outward?

- a) 0.13 b) 0.14 c) 0.15 d) 0.16 e) 0.17